

MATH I – Week 3

Fall 2026

Limits (of functions)

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Week 3 - READING: Sections 2.1, 2.2, 2.5 from textbook

emphasis on 2.2

Chapter Outline

- 2.1 A Preview of Calculus
- 2.2 The Limit of a Function
- 2.3 The Limit Laws
- 2.4 Continuity
- 2.5 The Precise Definition of a Limit



Introduction

Science fiction writers often imagine spaceships that can travel to far-off planets in distant galaxies. However, back in 1905, Albert Einstein showed that a limit exists to how fast any object can travel. The problem is that the faster an object moves, the more mass it attains (in the form of energy), according to the equation

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}},$$

where m_0 is the object's mass at rest, v is its speed, and c is the speed of light. What is this speed limit? (We explore this problem further in [Example 2.12.](#))

The idea of a limit is central to all of calculus. We begin this chapter by examining why limits are so important. Then, we

The Tangent Problem and Differential Calculus

Rate of change is one of the most critical concepts in calculus. We begin our investigation of rates of change by looking at the graphs of the three lines $f(x) = -2x - 3$, $g(x) = \frac{1}{2}x + 1$, and $h(x) = 2$, shown in [Figure 2.2](#).

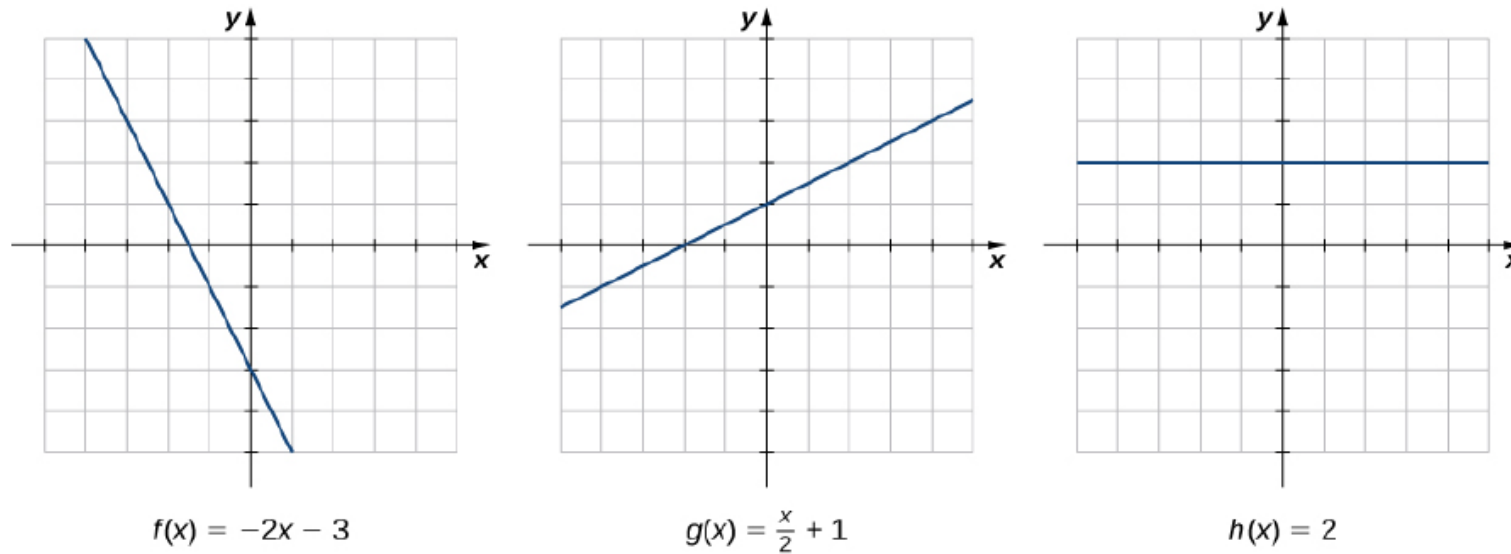


Figure 2.2 The rate of change of a linear function is constant in each of these three graphs, with the constant determined by the slope.

As we move from left to right along the graph of $f(x) = -2x - 3$, we see that the graph decreases at a constant rate.

What about
rate of change

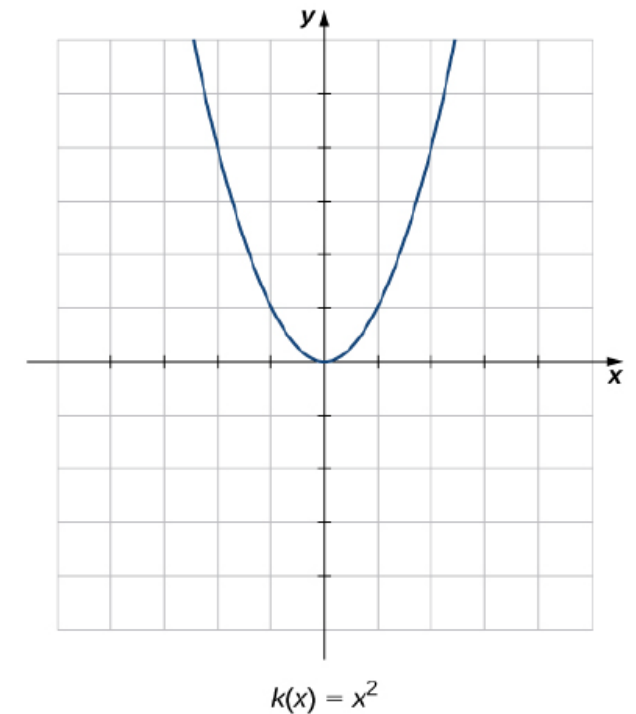


Figure 2.3 The function $k(x) = x^2$ does not have a constant rate of change.

The secant line *approximating the rate of change...*

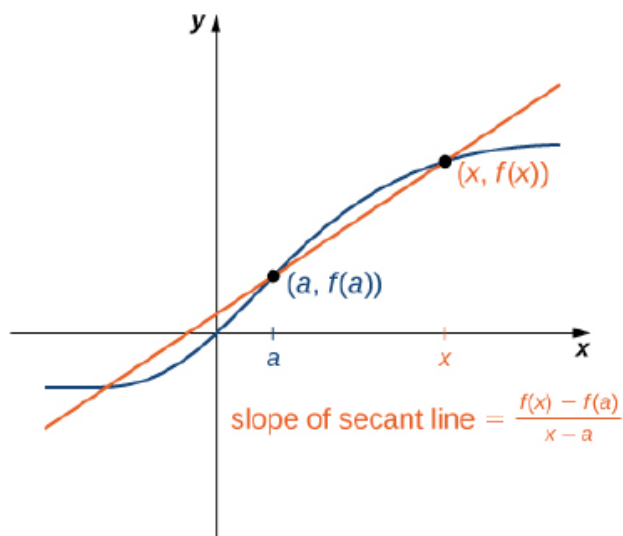


Figure 2.4 The slope of a secant line through a point $(a, f(a))$ estimates the rate of change of the function at the point $(a, f(a))$.

We formally define a secant line as follows:

Definition

The **secant** to the function $f(x)$ through the points $(a, f(a))$ and $(x, f(x))$ is the line passing through these points. Its slope is given by

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a}. \quad (2.1)$$

Secant line approaching the tangent of $f(x)$ at point a

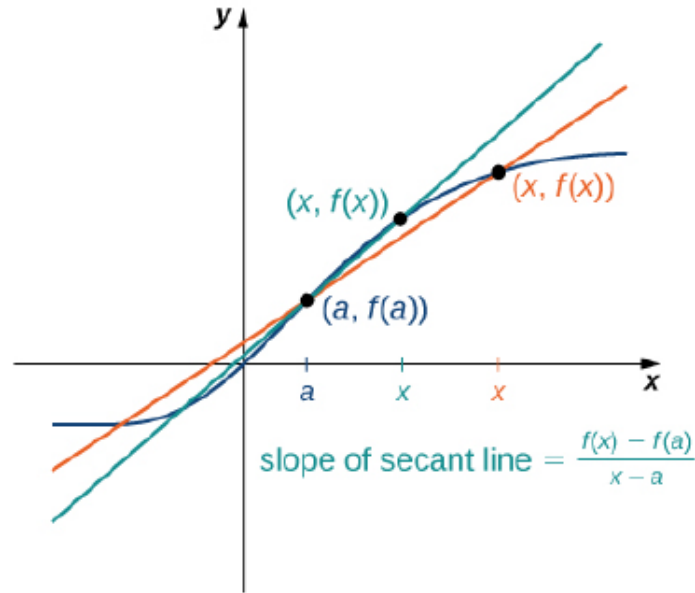


Figure 2.5 As x gets closer to a , the slope of the secant line becomes a better approximation to the rate of change of the function $f(x)$ at a .

The secant lines themselves approach a line that is called the **tangent** to the function $f(x)$ at a (Figure 2.6). The slope of the tangent line to the graph at a measures the rate of change of the function at a . This value also represents the derivative of the function $f(x)$ at a , or the rate of change of the function at a . This derivative is denoted by $f'(a)$.

Differential calculus is the field of calculus concerned with the study of derivatives and their applications.

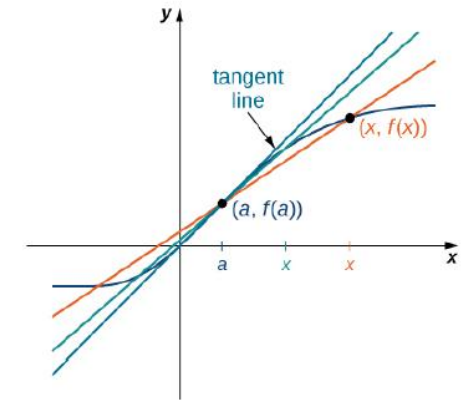


Figure 2.6 Solving the Tangent Problem: As x approaches a , the secant lines approach the tangent line.

Definition

Let $s(t)$ be the position of an object moving along a coordinate axis at time t . The **average velocity** of the object over a time interval $[a, t]$ where $a < t$ (or $[t, a]$ if $t < a$) is

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}. \quad (2.2)$$

As t is chosen closer to a , the average velocity becomes closer to the instantaneous velocity. Note that finding the average velocity of a position function over a time interval is essentially the same as finding the slope of a secant line to a function. Furthermore, to find the slope of a tangent line at a point a , we let the x -values approach a in the slope of the secant line. Similarly, to find the instantaneous velocity at time a , we let the t -values approach a in the average velocity. This process of letting x or t approach a in an expression is called taking a **limit**. Thus, we may define the **instantaneous velocity** as follows.

Definition

For a position function $s(t)$, the **instantaneous velocity** at a time $t = a$ is the value that the average velocities approach on intervals of the form $[a, t]$ and $[t, a]$ as the values of t become closer to a , provided such a value exists.

An example:
average velocity
instantaneous velocity

[Example 2.2](#) illustrates this concept of limits and average velocity.

EXAMPLE 2.2

Finding Average Velocity

A rock is dropped from a height of 64 ft. It is determined that its height (in feet) above ground t seconds later (for $0 \leq t \leq 2$) is $s(t) = -16t^2 + 64$. Find the average velocity of the rock over each of the given time intervals. Use this information to guess the instantaneous velocity of the rock at time $t = 0.5$.

- a. $[0.49, 0.5]$
- b. $[0.5, 0.51]$

✓ Solution

Substitute the data into the formula for the definition of average velocity.

$$\begin{aligned} \text{a. } v_{\text{ave}} &= \frac{s(0.5) - s(0.49)}{0.5 - 0.49} = -15.84 \\ \text{b. } v_{\text{ave}} &= \frac{s(0.51) - s(0.5)}{0.51 - 0.5} = -16.16 \end{aligned}$$

The instantaneous velocity is somewhere between -15.84 and -16.16 ft/sec. A good guess might be -16 ft/sec.

Limits

We begin our exploration of limits by taking a look at the graphs of the functions

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad g(x) = \frac{|x - 2|}{x - 2}, \quad \text{and} \quad h(x) = \frac{1}{(x - 2)^2},$$

which are shown in [Figure 2.12](#). In particular, let's focus our attention on the behavior of each graph at and around $x = 2$.

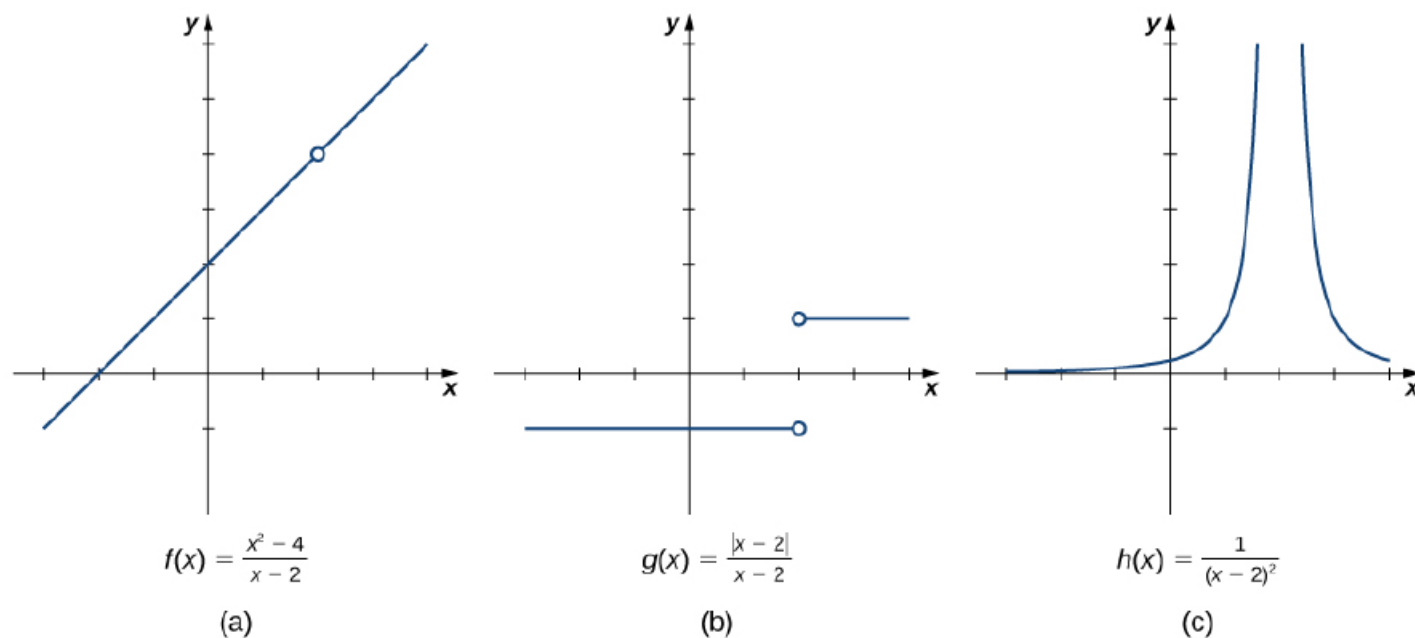


Figure 2.12 These graphs show the behavior of three different functions around $x = 2$.

Intuitive Definition of a Limit

Let's first take a closer look at how the function $f(x) = (x^2 - 4)/(x - 2)$ behaves around $x = 2$ in [Figure 2.12](#). As the values of x approach 2 from either side of 2, the values of $y = f(x)$ approach 4. Mathematically, we say that the limit of $f(x)$ as x approaches 2 is 4. Symbolically, we express this limit as

$$\lim_{x \rightarrow 2} f(x) = 4.$$

From this very brief informal look at one limit, let's start to develop an **intuitive definition of the limit**. We can think of the limit of a function at a number a as being the one real number L that the functional values approach as the x -values approach a , provided such a real number L exists. Stated more carefully, we have the following definition:

Definition

Let $f(x)$ be a function defined at all values in an open interval containing a , with the possible exception of a itself, and let L be a real number. If *all* values of the function $f(x)$ approach the real number L as the values of x ($\neq a$) approach the number a , then we say that the limit of $f(x)$ as x approaches a is L . (More succinct, as x gets closer to a , $f(x)$ gets closer and stays close to L .) Symbolically, we express this idea as

$$\lim_{x \rightarrow a} f(x) = L. \tag{2.3}$$

In-Class activity

EXAMPLE 2.6

Evaluating a Limit Using a Graph

For $g(x)$ shown in [Figure 2.15](#), evaluate $\lim_{x \rightarrow -1} g(x)$.

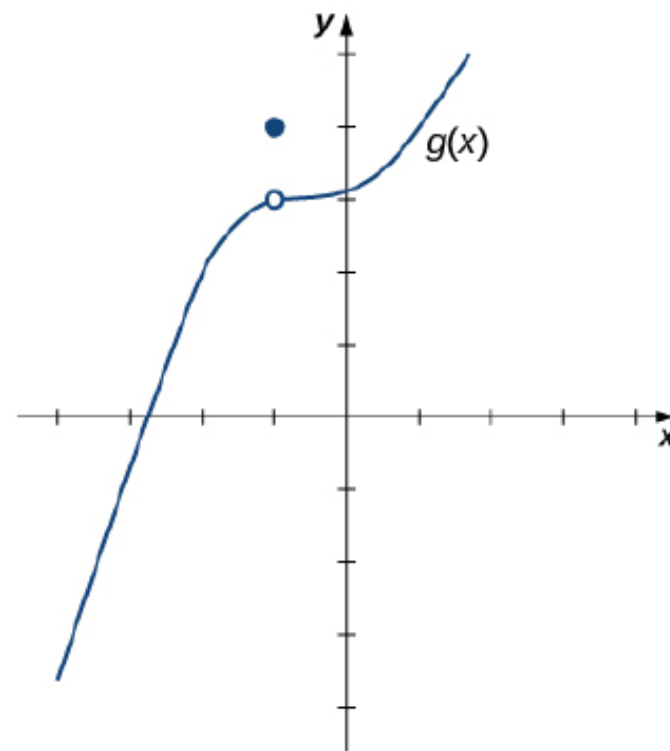


Figure 2.15 The graph of $g(x)$ includes one value not on a smooth curve.

In-Class activity

EXAMPLE 2.6

Evaluating a Limit Using a Graph

For $g(x)$ shown in [Figure 2.15](#), evaluate $\lim_{x \rightarrow -1} g(x)$.

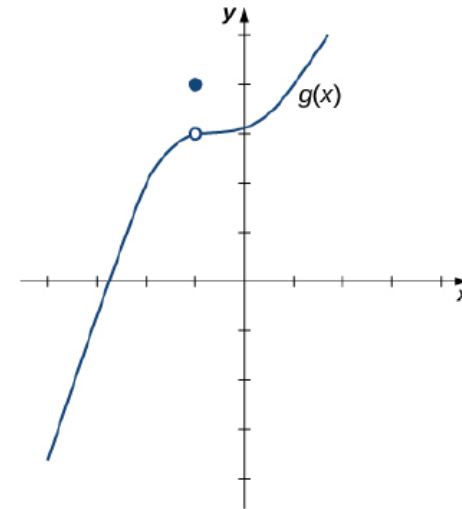


Figure 2.15 The graph of $g(x)$ includes one value not on a smooth curve.

✓ Solution

Despite the fact that $g(-1) = 4$, as the x -values approach -1 from either side, the $g(x)$ values approach 3 . Therefore, $\lim_{x \rightarrow -1} g(x) = 3$. Note that we can determine this limit without even knowing the algebraic expression of the function.

Example

x	$\frac{\sin x}{x}$	x	$\frac{\sin x}{x}$
-0.1	0.998334166468	0.1	0.998334166468
-0.01	0.999983333417	0.01	0.999983333417
-0.001	0.999999833333	0.001	0.999999833333
-0.0001	0.999999998333	0.0001	0.999999998333

Table 2.2 Table of Functional Values for $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

Note: The values in this table were obtained using a calculator and using all the places given in the calculator output.

As we read down each $\frac{(\sin x)}{x}$ column, we see that the values in each column appear to be approaching one. Thus, it is fairly reasonable to conclude that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. A calculator or computer-generated graph of $f(x) = \frac{(\sin x)}{x}$ would be similar to that shown in [Figure 2.13](#), and it confirms our estimate.

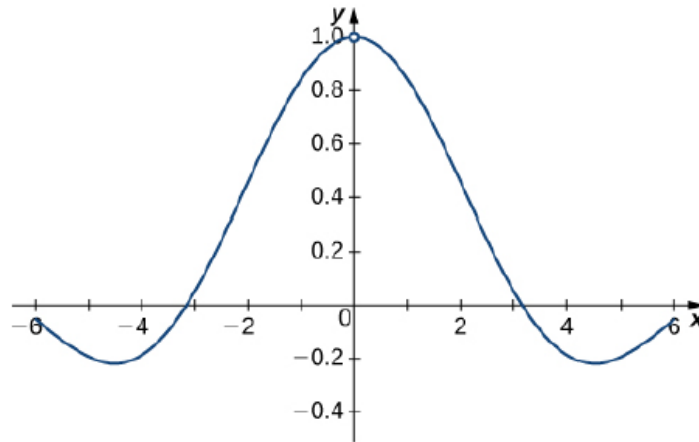


Figure 2.13 The graph of $f(x) = (\sin x)/x$ confirms the estimate from [Table 2.2](#).

The Existence of a Limit

As we consider the limit in the next example, keep in mind that for the limit of a function to exist at a point, the functional values must approach a single real-number value at that point. If the functional values do not approach a single value, then the limit does not exist.

EXAMPLE 2.7

Evaluating a Limit That Fails to Exist

Evaluate $\lim_{x \rightarrow 0} \sin(1/x)$ using a table of values.

✓ Solution

Table 2.5 lists values for the function $\sin(1/x)$ for the given values of x .

x	$\sin\left(\frac{1}{x}\right)$	x	$\sin\left(\frac{1}{x}\right)$
-0.1	0.544021110889	0.1	-0.544021110889
-0.01	0.50636564111	0.01	-0.50636564111
-0.001	-0.8268795405312	0.001	0.826879540532
-0.0001	0.305614388888	0.0001	-0.305614388888

Table 2.5 Table of Functional Values for $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$

After examining the table of functional values, we can see that the y -values do not seem to approach any one single value. It appears the limit does not exist. Before drawing this conclusion, let's take a more systematic approach. Take the following sequence of x -values approaching 0:

$$\frac{2}{\pi}, \frac{2}{3\pi}, \frac{2}{5\pi}, \frac{2}{7\pi}, \frac{2}{9\pi}, \frac{2}{11\pi}, \dots$$

The corresponding y -values are

$$1, -1, 1, -1, 1, -1, \dots$$

At this point we can indeed conclude that $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist. (Mathematicians frequently abbreviate “does not exist” as DNE. Thus, we would write $\lim_{x \rightarrow 0} \sin(1/x)$ DNE.) The graph of $f(x) = \sin(1/x)$ is shown in Figure 2.17 and it gives a clearer picture of the behavior of $\sin(1/x)$ as x approaches 0. You can see that $\sin(1/x)$ oscillates ever more wildly between -1 and 1 as x approaches 0.

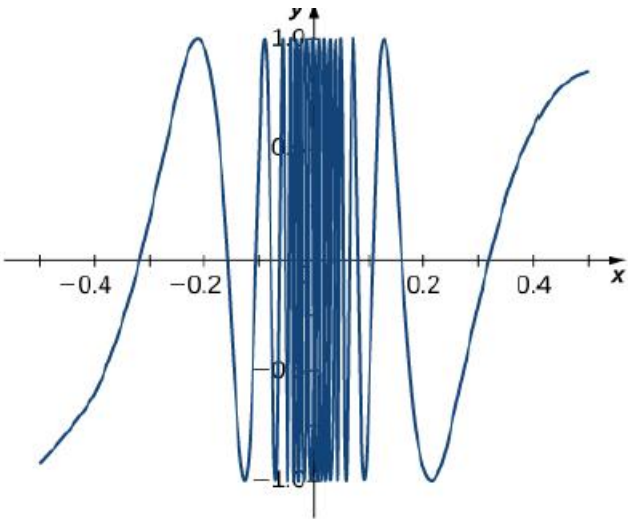


Figure 2.17 The graph of $f(x) = \sin(1/x)$ oscillates rapidly between -1 and 1 as x approaches 0.

One-sided limits

Definition

We define two types of **one-sided limits**.

Limit from the left: Let $f(x)$ be a function defined at all values in an open interval of the form (c, a) , and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x < a$) approach the number a , then we say that L is the limit of $f(x)$ as x approaches a from the left. Symbolically, we express this idea as

$$\lim_{x \rightarrow a^-} f(x) = L. \quad (2.6)$$

Limit from the right: Let $f(x)$ be a function defined at all values in an open interval of the form (a, c) , and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x > a$) approach the number a , then we say that L is the limit of $f(x)$ as x approaches a from the right. Symbolically, we express this idea as

$$\lim_{x \rightarrow a^+} f(x) = L. \quad (2.7)$$

In-Class activity

Evaluating One-Sided Limits

For the function $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits.

a. $\lim_{x \rightarrow 2^-} f(x)$

b. $\lim_{x \rightarrow 2^+} f(x)$

EXAMPLE 2.8

Evaluating One-Sided Limits

For the function $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits.

- a. $\lim_{x \rightarrow 2^-} f(x)$
- b. $\lim_{x \rightarrow 2^+} f(x)$

✓ Solution

We can use tables of functional values again [Table 2.6](#). Observe that for values of x less than 2, we use $f(x) = x + 1$ and for values of x greater than 2, we use $f(x) = x^2 - 4$.

x	$f(x) = x + 1$	x	$f(x) = x^2 - 4$
1.9	2.9	2.1	0.41
1.99	2.99	2.01	0.0401
1.999	2.999	2.001	0.004001
1.9999	2.9999	2.0001	0.00040001
1.99999	2.99999	2.00001	0.0000400001

Table 2.6 Table of Functional Values for

$$f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$$

Based on this table, we can conclude that a. $\lim_{x \rightarrow 2^-} f(x) = 3$ and b. $\lim_{x \rightarrow 2^+} f(x) = 0$. Therefore, the (two-sided) limit of

$f(x)$ does not exist at $x = 2$. [Figure 2.18](#) shows a graph of $f(x)$ and reinforces our conclusion about these limits.

Solution

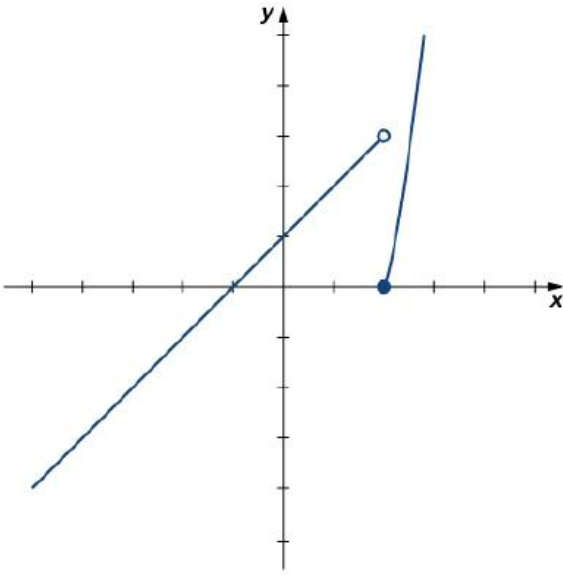


Figure 2.18 The graph of $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$ has a break at $x = 2$.

One-Sided and Two-Sided Limits

Theorem 2.2

Relating One-Sided and Two-Sided Limits

Let $f(x)$ be a function defined at all values in an open interval containing a , with the possible exception of a itself, and let L be a real number. Then,

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L.$$

Infinite limits

Definition

We define three types of **infinite limits**.

Infinite limits from the left: Let $f(x)$ be a function defined at all values in an open interval of the form (b, a) .

- i. If the values of $f(x)$ increase without bound as the values of x (where $x < a$) approach the number a , then we say that the limit as x approaches a from the left is positive infinity and we write

$$\lim_{x \rightarrow a^-} f(x) = +\infty. \quad (2.8)$$

- ii. If the values of $f(x)$ decrease without bound as the values of x (where $x < a$) approach the number a , then we say that the limit as x approaches a from the left is negative infinity and we write

$$\lim_{x \rightarrow a^-} f(x) = -\infty. \quad (2.9)$$

Infinite limits from the right: Let $f(x)$ be a function defined at all values in an open interval of the form (a, c) .

- i. If the values of $f(x)$ increase without bound as the values of x (where $x > a$) approach the number a , then we say that the limit as x approaches a from the right is positive infinity and we write

$$\lim_{x \rightarrow a^+} f(x) = +\infty. \quad (2.10)$$

- ii. If the values of $f(x)$ decrease without bound as the values of x (where $x > a$) approach the number a , then we say that the limit as x approaches a from the right is negative infinity and we write

$$\lim_{x \rightarrow a^+} f(x) = -\infty. \quad (2.11)$$

Two-sided infinite limit: Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a .

- i. If the values of $f(x)$ increase without bound as the values of x (where $x \neq a$) approach the number a , then we say that the limit as x approaches a is positive infinity and we write

$$\lim_{x \rightarrow a} f(x) = +\infty. \quad (2.12)$$

- ii. If the values of $f(x)$ decrease without bound as the values of x (where $x \neq a$) approach the number a , then we say that the limit as x approaches a is negative infinity and we write

$$\lim_{x \rightarrow a} f(x) = -\infty. \quad (2.13)$$

In-Class activity

Recognizing an Infinite Limit

Evaluate each of the following limits, if possible conclusion.

a. $\lim_{x \rightarrow 0^-} \frac{1}{x}$

b. $\lim_{x \rightarrow 0^+} \frac{1}{x}$

c. $\lim_{x \rightarrow 0} \frac{1}{x}$

Solution

- a. The values of $1/x$ decrease without bound as x approaches 0 from the left. We conclude that

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

- b. The values of $1/x$ increase without bound as x approaches 0 from the right. We conclude that

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty.$$

- c. Since $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$ and $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$ have different values, we conclude that

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ DNE.}$$

The graph of $f(x) = 1/x$ in [Figure 2.19](#) confirms these conclusions.

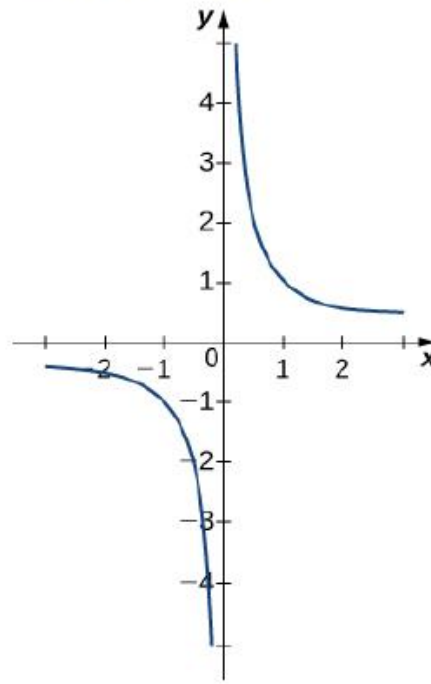


Figure 2.19 The graph of $f(x) = 1/x$ confirms that the limit as x approaches 0 does not exist.

In-Class activity

Evaluate each of the following limits, if possible

a. $\lim_{x \rightarrow 0^-} \frac{1}{x^2}$

b. $\lim_{x \rightarrow 0^+} \frac{1}{x^2}$

c. $\lim_{x \rightarrow 0} \frac{1}{x^2}$

It is useful to point out that functions of the form $f(x) = 1/(x - a)^n$, where n is a positive integer, have infinite limits as x approaches a from either the left or right (Figure 2.20). These limits are summarized in [Infinite Limits from Positive Integers](#).

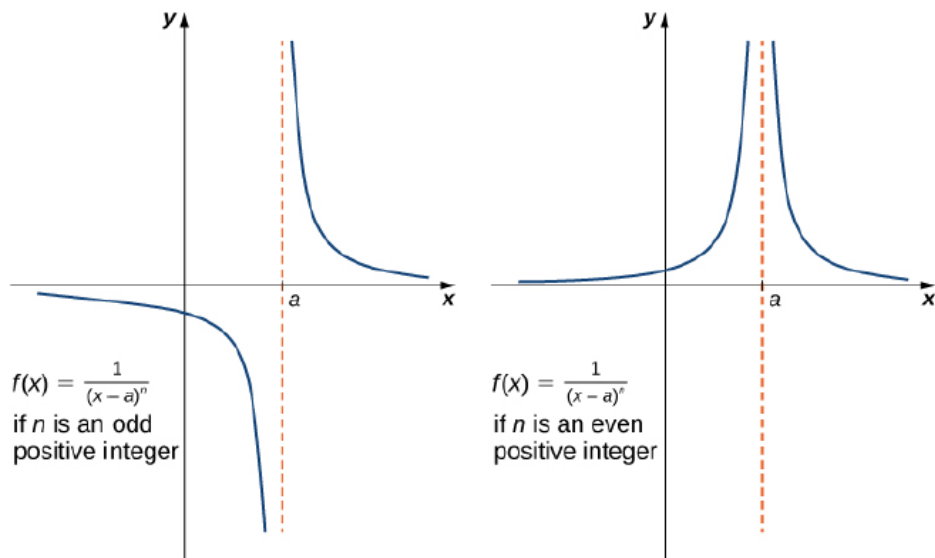


Figure 2.20 The function $f(x) = 1/(x - a)^n$ has infinite limits at a .

Theorem 2.3

Infinite Limits from Positive Integers

If n is a positive even integer, then

$$\lim_{x \rightarrow a} \frac{1}{(x - a)^n} = +\infty.$$

If n is a positive odd integer, then

$$\lim_{x \rightarrow a^+} \frac{1}{(x - a)^n} = +\infty$$

and

$$\lim_{x \rightarrow a^-} \frac{1}{(x - a)^n} = -\infty.$$

Solution

We should also point out that in the graphs of $f(x) = 1/(x - a)^n$, points on the graph having x -coordinates very near to a are very close to the vertical line $x = a$. That is, as x approaches a , the points on the graph of $f(x)$ are closer to the line $x = a$. The line $x = a$ is called a **vertical asymptote** of the graph. We formally define a vertical asymptote as follows:

Definition

Let $f(x)$ be a function. If any of the following conditions hold, then the line $x = a$ is a **vertical asymptote** of $f(x)$.

$$\lim_{x \rightarrow a^-} f(x) = +\infty \text{ or } -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = +\infty \text{ or } -\infty$$

or

$$\lim_{x \rightarrow a} f(x) = +\infty \text{ or } -\infty$$

Example

EXAMPLE 2.10

Finding a Vertical Asymptote

Evaluate each of the following limits using [Infinite Limits from Positive Integers](#). Identify any vertical asymptotes of the function $f(x) = 1/(x + 3)^4$.

- a. $\lim_{x \rightarrow -3^-} \frac{1}{(x+3)^4}$
- b. $\lim_{x \rightarrow -3^+} \frac{1}{(x+3)^4}$
- c. $\lim_{x \rightarrow -3} \frac{1}{(x+3)^4}$

✓ Solution

We can use [Infinite Limits from Positive Integers](#) directly.

- a. $\lim_{x \rightarrow -3^-} \frac{1}{(x+3)^4} = +\infty$
- b. $\lim_{x \rightarrow -3^+} \frac{1}{(x+3)^4} = +\infty$
- c. $\lim_{x \rightarrow -3} \frac{1}{(x+3)^4} = +\infty$

The function $f(x) = 1/(x + 3)^4$ has a vertical asymptote of $x = -3$.

In-Class activity

EXAMPLE 2.11

Behavior of a Function at Different Points

Use the graph of $f(x)$ in [Figure 2.21](#) to determine each of the following values:

- a. $\lim_{x \rightarrow -4^-} f(x)$; $\lim_{x \rightarrow -4^+} f(x)$; $\lim_{x \rightarrow -4} f(x)$; $f(-4)$
- b. $\lim_{x \rightarrow -2^-} f(x)$; $\lim_{x \rightarrow -2^+} f(x)$; $\lim_{x \rightarrow -2} f(x)$; $f(-2)$
- c. $\lim_{x \rightarrow 1^-} f(x)$; $\lim_{x \rightarrow 1^+} f(x)$; $\lim_{x \rightarrow 1} f(x)$; $f(1)$
- d. $\lim_{x \rightarrow 3^-} f(x)$; $\lim_{x \rightarrow 3^+} f(x)$; $\lim_{x \rightarrow 3} f(x)$; $f(3)$

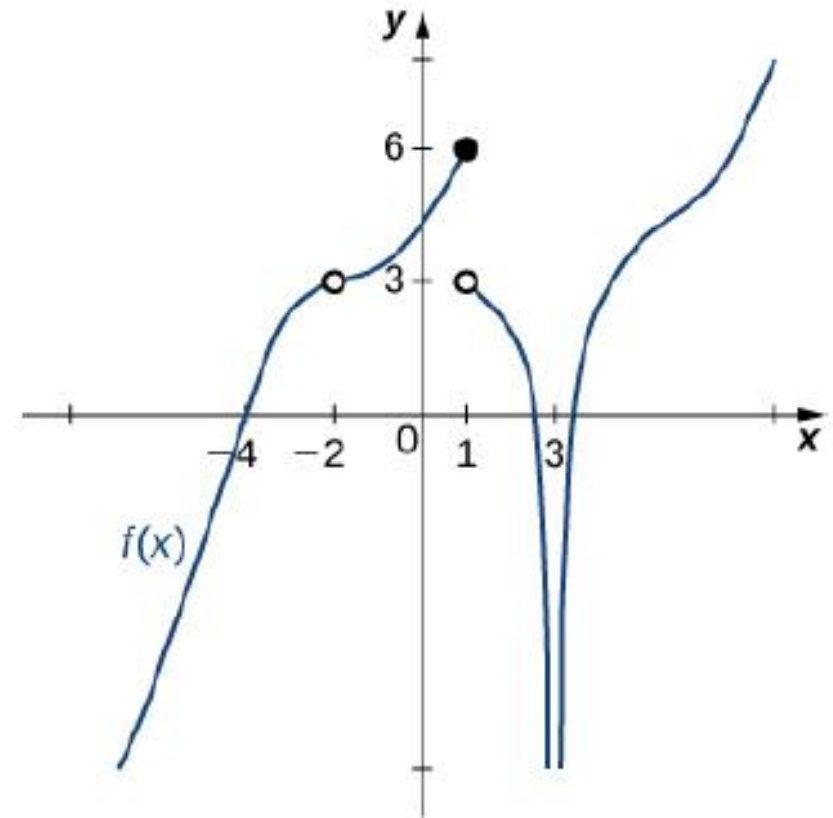


Figure 2.21 The graph shows $f(x)$.

EXAMPLE 2.11

Behavior of a Function at Different Points

Use the graph of $f(x)$ in [Figure 2.21](#) to determine each of the following values:

- a. $\lim_{x \rightarrow -4^-} f(x)$; $\lim_{x \rightarrow -4^+} f(x)$; $\lim_{x \rightarrow -4} f(x)$; $f(-4)$
- b. $\lim_{x \rightarrow -2^-} f(x)$; $\lim_{x \rightarrow -2^+} f(x)$; $\lim_{x \rightarrow -2} f(x)$; $f(-2)$
- c. $\lim_{x \rightarrow 1^-} f(x)$; $\lim_{x \rightarrow 1^+} f(x)$; $\lim_{x \rightarrow 1} f(x)$; $f(1)$
- d. $\lim_{x \rightarrow 3^-} f(x)$; $\lim_{x \rightarrow 3^+} f(x)$; $\lim_{x \rightarrow 3} f(x)$; $f(3)$

✓ Solution

Using [Infinite Limits from Positive Integers](#) and the graph for reference, we arrive at the following values:

- a. $\lim_{x \rightarrow -4^-} f(x) = 0$; $\lim_{x \rightarrow -4^+} f(x) = 0$; $\lim_{x \rightarrow -4} f(x) = 0$; $f(-4) = 0$
- b. $\lim_{x \rightarrow -2^-} f(x) = 3$; $\lim_{x \rightarrow -2^+} f(x) = 3$; $\lim_{x \rightarrow -2} f(x) = 3$; $f(-2)$ is undefined
- c. $\lim_{x \rightarrow 1^-} f(x) = 6$; $\lim_{x \rightarrow 1^+} f(x) = 3$; $\lim_{x \rightarrow 1} f(x)$ DNE; $f(1) = 6$
- d. $\lim_{x \rightarrow 3^-} f(x) = -\infty$; $\lim_{x \rightarrow 3^+} f(x) = -\infty$; $\lim_{x \rightarrow 3} f(x) = -\infty$; $f(3)$ is undefined

In-Class activity

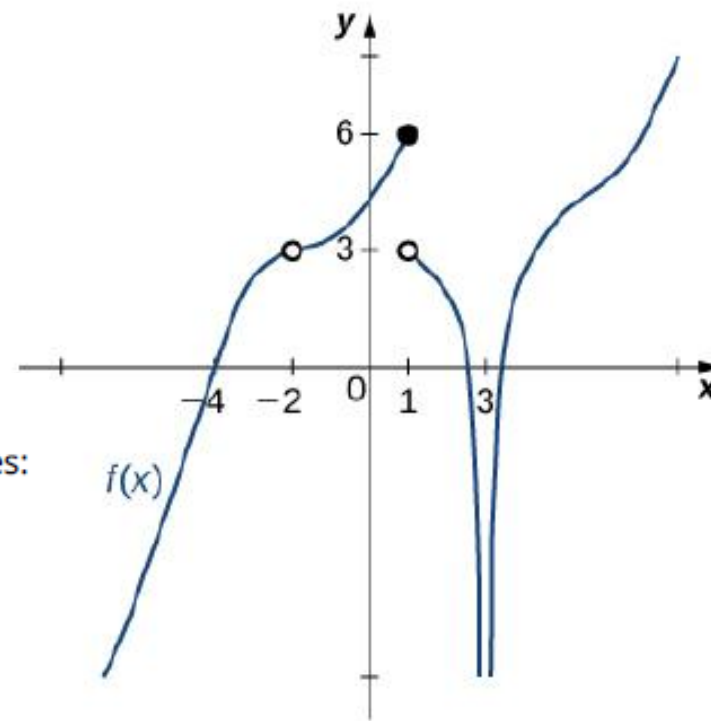


Figure 2.21 The graph shows $f(x)$.

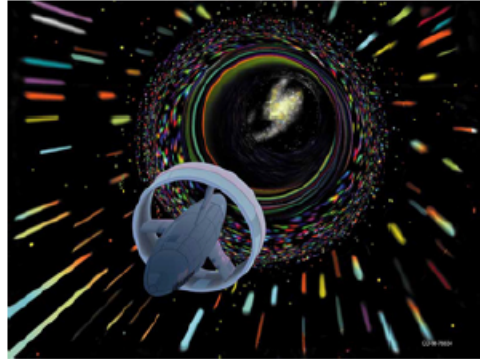


Figure 2.22 (credit: NASA)

...what happens if you move
very very fast?

In the chapter opener we mentioned briefly how Albert Einstein showed that a limit exists to how fast any object can travel. Given Einstein's equation for the mass of a moving object, what is the value of this bound?

✓ **Solution**

Our starting point is Einstein's equation for the mass of a moving object,

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}},$$

where m_0 is the object's mass at rest, v is its speed, and c is the speed of light. To see how the mass changes at high speeds, we can graph the ratio of masses m/m_0 as a function of the ratio of speeds, v/c (Figure 2.23).

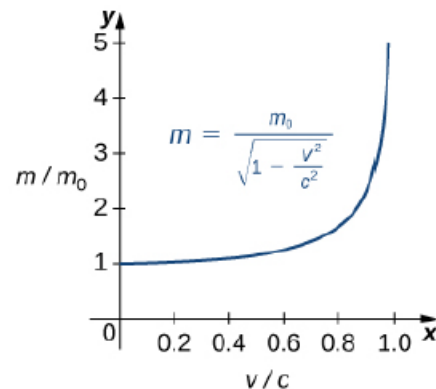


Figure 2.23 This graph shows the ratio of masses as a function of the ratio of speeds in Einstein's equation for the mass of a moving object.

The precise definition of the limit

Quantifying Closeness

Before stating the formal definition of a limit, we must introduce a few preliminary ideas. Recall that the distance between two points a and b on a number line is given by $|a - b|$.

- The statement $|f(x) - L| < \varepsilon$ may be interpreted as: *The distance between $f(x)$ and L is less than ε .*
- The statement $0 < |x - a| < \delta$ may be interpreted as: *$x \neq a$ and the distance between x and a is less than δ .*

It is also important to look at the following equivalences for absolute value:

- The statement $|f(x) - L| < \varepsilon$ is equivalent to the statement $L - \varepsilon < f(x) < L + \varepsilon$.
- The statement $0 < |x - a| < \delta$ is equivalent to the statement $a - \delta < x < a + \delta$ and $x \neq a$.

The ε - δ definition of the limit

With these clarifications, we can state the formal **epsilon-delta definition of the limit**.

Definition

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a . Let L be a real number. Then

$$\lim_{x \rightarrow a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists a $\delta > 0$, such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

This definition may seem rather complex from a mathematical point of view, but it becomes easier to understand if we break it down phrase by phrase. The statement itself involves something called a *universal quantifier* (for every $\varepsilon > 0$), an *existential quantifier* (there exists a $\delta > 0$), and, last, a *conditional statement* (if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$). Let's take a look at [Table 2.9](#), which breaks down the definition and translates each part.

Definition	Translation
1. For every $\varepsilon > 0$,	1. For every positive distance ε from L ,
2. there exists a $\delta > 0$,	2. There is a positive distance δ from a ,
3. such that	3. such that
4. if $0 < x - a < \delta$, then $ f(x) - L < \varepsilon$.	4. if x is closer than δ to a and $x \neq a$, then $f(x)$ is closer than ε to L .

Table 2.9 Translation of the Epsilon-Delta Definition of the Limit

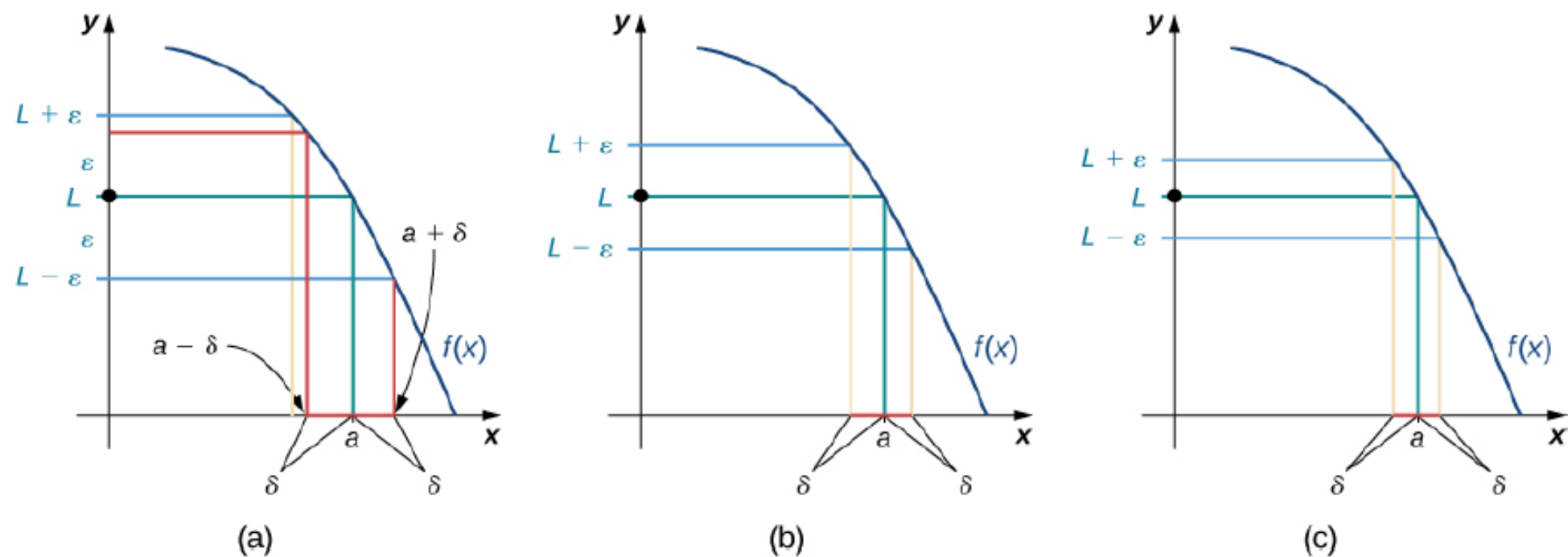


Figure 2.39 These graphs show possible values of δ , given successively smaller choices of ϵ .